One Score Cannot Serve Every Decision: Criterion Conflict and Failure Composition in Segmentation Quality

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Code and derived results

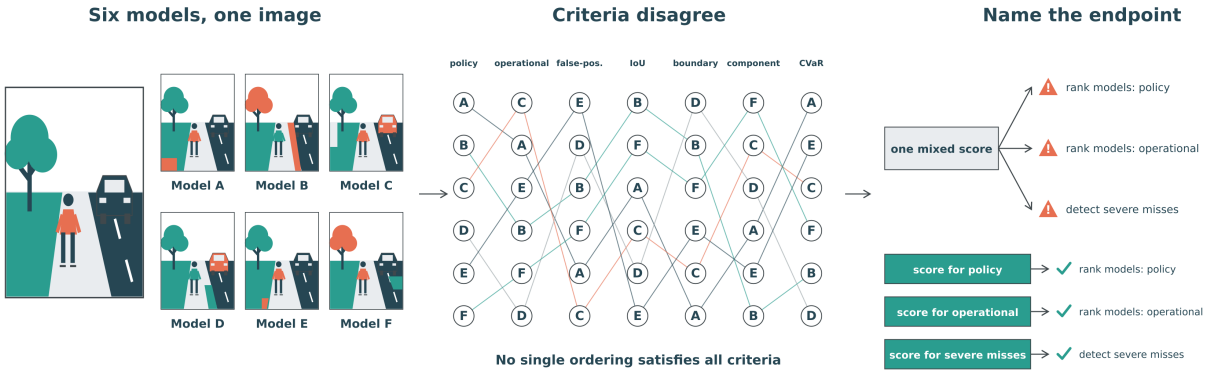


Figure 1. Six models segment the same image (left; schematic illustration). Seven declared quality criteria rank them differently (middle; illustrative orderings), so no single ordering satisfies all of them: on our eight panels every scalar ranking contradicts at least 0.106–0.189 of the non-tied criterion preferences. Reusing one mixed score for every decision mismatches the endpoints; naming the endpoint and scoring it with its own frozen formula improves each non-trivial endpoint on every panel (right).

Abstract

Segmentation quality scores serve different decisions, such as ranking models and flagging failed predictions, yet are usually reported as if one number served them all. We test this on eight six-model panels in three domains: general scenes (COCO, PASCAL VOC, and three ADE20K panels including recent architectures), street scenes (Cityscapes, KITTI), and abdominal CT (AMOS). For seven declared criteria, every scalar ordering of the models must contradict 0.106–0.189 of the non-tied criterion preferences, and exhaustive search over all 720 orderings shows this bound is essentially exact. Scoring each declared endpoint with its own frozen formula instead of one mixed score improves policy-rank agreement, operational-rank agreement, and severe-miss AUROC on all eight panels; all 24 contrasts are significant after Holm correction over 40 tests. A single failure score is more fragile. Selected on general scenes, it beats the strongest single metric there, but not on street scenes and CT, where high-error failures make up at most 6.4% of failures and severe misses dominate. Re-selecting it on Cityscapes training data brings it level with the strongest single metric on three held-out panels, including CT (gains of +0.114, +0.184, and +0.017 AUROC over the general-scene selection). Quality evaluation should therefore name its endpoint and match its failure score to the failure composition of the target domain.

1. Introduction

Segmentation metrics summarize different properties of a prediction. Mean intersection over union (mIoU) measures region overlap; Boundary IoU emphasizes contours; tail statistics emphasize poorly covered components; and false-positive scores penalize unsupported predictions. A score that separates failed from non-failed images need not rank models in the order a declared policy or operational utility

would. Treating these uses as interchangeable obscures the decision a metric is meant to support, and the problem is sharper outside the general-scene benchmarks on which most scores are developed.

We study the issue through pairwise preferences over complete six-model image groups (Fig. 1). For each image and model pair, seven criteria may agree, tie, or disagree. The minority count among non-tied preferences lower-bounds the disagreement of any single ordering. Across eight panels in three domains the bound is 0.106–0.189, and an exhaustive search over all 720 strict orderings of each image shows it is essentially attained: the exact minimum exceeds it by at most 0.0023. In social-choice terms, this exact minimum is the Kemeny–Young optimal disagreement of the criteria viewed as voters [21], a quantitative complement to Arrow's impossibility theorem [1]. No scalar score, however it is built, can order these models without contradicting this share of the declared preferences.

Given an unavoidable conflict, we evaluate a simple protocol: select one score formula per endpoint on COCO and VOC, freeze the formulas, and score each endpoint with its own formula once the endpoint is declared. Against reuse of a single mixed-failure score, this improves rank agreement and severe-miss detection on all eight panels, including recent ADE20K architectures, street scenes, and CT, none of which was used for selection.

A single failure score behaves differently. Selected on general scenes, the mixed score is the best failure detector there, but on street scenes and CT it trails a single coverage or tail metric. Those domains have a different failure composition: high-error failures are rare and nearly every failure is a severe miss, often a small distant pedestrian in street scenes. Re-selecting the failure score on Cityscapes training data, whose failures share that composition, restores it on three held-out panels, including CT.

Our contributions are: (1) an empirical, essentially exact bound on scalar-ranking disagreement among declared segmentation criteria, computed as the Kemeny–Young optimum of the criteria viewed as voters, on eight panels in three domains; (2) a cross-domain evaluation of frozen endpoint-specific scoring, which improves every non-trivial endpoint on every panel; and (3) evidence that a single failure score transfers only between domains with similar failure composition, and that selecting it on data with matching composition restores it. All results are conditional on the declared criteria, panels, and endpoints; they support naming the decision endpoint and its target failure composition when reporting segmentation quality, not a new universal metric.

2. Related Work

Segmentation metrics and problem-aware validation. Mean intersection over union is widely used for semantic segmentation, but it compresses class- and pixel-level error structure into one aggregate. Boundary IoU is designed to increase sensitivity to contour errors, particularly where standard region overlap is insensitive to object-scale boundary deviations [7]. More broadly, validation metrics can produce misleading comparisons through implementation, aggregation, ranking, or interpretation choices. Metrics Reloaded recommends selecting complementary measures from an explicit description of the target problem, and its companion analysis organizes recurrent metric-related pitfalls [26, 27]. We operationalize one such concern: different correctly computed criteria can induce incompatible model preferences. Our finite-panel bound quantifies that incompatibility for the criteria declared here; it does not identify a universally optimal metric set.

Rank aggregation and social choice. Combining several rankings into one is a classical problem. Arrow's impossibility theorem rules out aggregation rules that satisfy a few natural axioms simultaneously [1], and the Kemeny–Young rule selects the ranking that minimizes total pairwise disagreement with the voters [21, 33]; computing it is NP-hard in general [3] and has been studied for combining search engine rankings [12]. We treat segmentation quality criteria as voters over models and use the Kemeny objective not to aggregate them, but to measure how much disagreement any single quality score must incur.

Reliability and uncertainty in segmentation. Semantic-segmentation reliability includes distinct tasks such as robustness, calibration, misclassification detection, and out-of-distribution detection. A broad architecture study found that architectural progress on robustness does not imply uniform improvement in uncertainty-based reliability [11]. ValUES advocates controlled validation across uncertainty applications and emphasizes that score aggregation is itself a consequential design choice [20]. Our scope is narrower: we compare fixed per-image quality scores for model ranking and failure discrimination. We do not estimate pixelwise predictive calibration, detect out-of-distribution inputs, or claim that these tasks are interchangeable.

Safety- and relevance-aware evaluation. Driving-segmentation metrics can account for spatial error structure [9]. Relevance-weighted IoU emphasizes corner-case pixels [4], while Safe mIoU represents class-confusion costs through a hierarchy [29]. These approaches make task priorities explicit. We similarly declare severity weights and operational thresholds, but treat them as experimental assumptions rather than validated harm models. We report policy agreement, operational agreement, severe misses, and false positives separately so that one endpoint cannot stand in for all others.

Tail risk, selection, and endpoint choice. Conditional value at risk (CVaR) summarizes the worst part of a loss distribution [28]; minimum component coverage provides an extreme component-level statistic. Selective-prediction methods instead evaluate risk as a function of retained coverage and can provide guarantees under their assumptions [14]. We evaluate neither abstention nor risk–coverage guarantees. Our question is whether failure-detection discrimination and model-ranking agreement can be treated as the same endpoint. The results show why that implication must be tested rather than assumed: a mixed-failure AUROC gain does not guarantee better policy or operational ranking.

Architectural diversity. The evaluated panels include FCN [25], DeepLabV3 [6], MobileNetV3 [17], SegFormer [32], UPerNet [31], ConvNeXt [24], Swin [23], BEiT [2], and Mask2Former [8]. This permits a limited cross-architecture analysis, but each panel contains only six evaluated checkpoints. It is not a broad sample of the architecture space.

3. Evaluation Framework

3.1. Component scores and endpoint targets

For image x and model m, let eligible target component i have coverage $c_i \in [0,1]$, provenance $p_i \in [0,1]$, and externally assigned severity $s_i > 0$. The candidate score set includes foreground IoU, Boundary IoU, minimum component coverage, severity-weighted arithmetic and harmonic means, CVaR quality, false-positive quality $q_{\mathrm{fp}} = 1 - r_{\mathrm{fp}}$, and the geometric component score

$$q_{\mathrm{comp}} = \exp\left(\sum_i \frac{s_i^\alpha}{\sum_j s_j^\alpha} \log[\epsilon + (1-\epsilon)c_i p_i]\right)(1-r_{\mathrm{fp}})^\gamma. \tag{1}$$

The implementation sets $\alpha = 2$, $\epsilon = 10^{-6}$, and $\gamma = 1$. Scores are oriented so that larger values indicate better quality. This score is one candidate in the comparison, not a universal recommendation.

The declared policy utility is one minus the largest severity-normalized component risk. Operational utility is the minimum of policy utility and q_{fp}. An image is labelled as containing a severe miss if an eligible critical component has coverage below 0.5. A high-false-positive image has $r_{\mathrm{fp}} \geq 0.25$. Their union defines the primary mixed-failure endpoint. These thresholds and severity weights are experimental operationalizations, not causal labels of harm or validated deployment costs.

All targets and score components are computed against reference annotations; these are retrospective, annotation-dependent quality assessments, not annotation-free online reliability estimates. The false-positive endpoint is particularly direct: $h_{\mathrm{fp}} = 1 - r_{\mathrm{fp}}$, so its error score is $1 - h_{\mathrm{fp}} = r_{\mathrm{fp}}$, while its label is $\mathbb{1}[r_{\mathrm{fp}} \geq 0.25]$. Whenever both label classes occur, this construction yields AUROC 1 by definition. We treat that result as a check of alignment with a directly constructed endpoint, not as evidence of general failure-detection ability.

Policy and operational targets are evaluated as rankings of six models within each image: we compute Spearman rank correlation for each image and average across images with non-degenerate ranks. Severe-miss, false-positive, and mixed-failure targets are binary image labels evaluated with AUROC using $1 - h_e$ as the error score. Rank agreement and AUROC therefore estimate different properties. Neither is a calibration measure, a deployment threshold, or a guarantee of utility after model selection.

3.2. Empirical pairwise-conflict bound

Let K criteria score all M models on image x. For model pair $p = (a, b)$, criterion k gives preference $y_{xpk} \in \{-1, 0, +1\}$. After excluding ties, let n_{xp}^+ and n_{xp}^- count preferences in each direction. Any single ordering must choose one direction for p and thus disagree with at least $\min(n_{xp}^+, n_{xp}^-)$ declared preferences. We aggregate the unavoidable minority count across image–pair instances:

$$B = \frac{\sum_{x,p} \min(n_{xp}^+, n_{xp}^-)}{\sum_{x,p}(n_{xp}^+ + n_{xp}^-)}. \tag{2}$$

Criterion ties are defined by exact numerical equality; the implementation applies no tolerance. The pairwise sum is a lower bound and need not be attained by a transitive total ordering. Its denominator is the number of non-tied criterion votes, not the number of images or deployment decisions. The bound concerns complete strict model rankings. A scalar score that ties models therefore requires a deterministic tie break; unresolved scalar ties are not treated as a separate outcome. This is a preference-weighted empirical lower bound for the finite criterion set, images, and model panel. It is not the fraction of images with differing rankings, a fitted-score error, or a theorem about unobserved criteria or models. The primary estimate excludes ties; sensitivity analyses count ties as agreement or split them between directions. Confidence intervals resample images while retaining each image's complete model group, preserving within-image dependence among model-pair comparisons.

Because a scalar score can realize any strict ordering of the models on an image, the least disagreement attainable by any scalar is obtained by enumerating all $M!$ orderings per image. With $M = 6$ we enumerate the 720 orderings, count the non-tied criterion votes each one contradicts, take the per-image minimum, and aggregate as in Eq. 2. We denote the result $B_{\mathrm{exact}} \geq B$; the gap measures how much intransitivity across model pairs adds to the pairwise bound.

Social-choice view. Treating each criterion as a voter that ranks the models (abstaining on ties), B_{exact} is the normal-

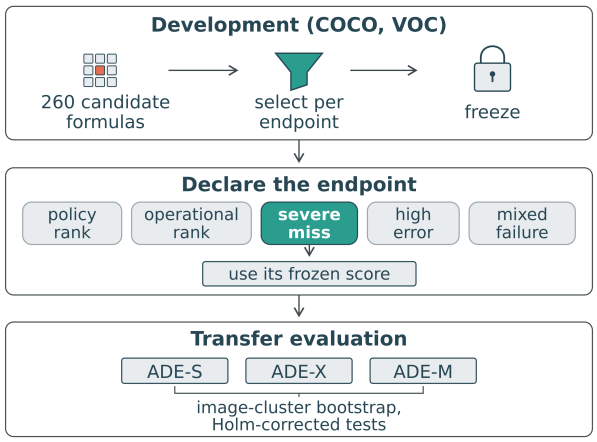


Figure 2. Protocol. One score formula per endpoint is selected from 260 candidates on COCO and VOC and frozen. After an endpoint is declared, its frozen score is applied unchanged to the transfer panels (ADE20K panels shown; street-scene and CT panels follow the same protocol), with image-cluster bootstrap intervals and Holm-corrected tests.

ized score of a Kemeny–Young optimal ranking [21, 33]: the least total pairwise disagreement achievable by any strict ranking. B is the classical pairwise lower bound on that score. It is attained on an image exactly when some strict ordering agrees with every strict pairwise majority, so a small gap means majority cycles are rare and the disagreement is pairwise. Kemeny aggregation is NP-hard in general [3, 12], but exhaustive search is trivial for six models. Arrow's theorem [1] shows that no rule can aggregate rankings of three or more alternatives while meeting a small set of natural conditions, but it does not say how much disagreement any particular aggregate must incur. B and B_{exact} provide that quantity for the observed preference profiles: every scalar quality score, being one strict ranking per image, must contradict at least this share of the declared criteria.

3.3. Frozen endpoint-specific scores

We enumerate single components and convex pairs on a fixed weight grid. The development selection uses COCO and VOC and requires cross-dataset consensus; the selected formulas are then frozen:

$$h_{\text{policy}} = q_{\text{CVaR}}, \qquad h_{\text{oper}} = .1q_{\text{CVaR}} + .9q_{\text{comp}}, \tag{3}$$

$$h_{\text{severe}} = .9q_{\text{CVaR}} + .1q_{\text{fp}}, \qquad h_{\text{fp}} = q_{\text{fp}}, \tag{4}$$

$$h_{\text{mixed}} = .2q_{\text{comp}} + .8q_{\text{fp}}. \tag{5}$$

For endpoint e, routing selects h_e after e has been declared. The baseline reuses h_{mixed} for all endpoints. This estimates the effect of assigning these frozen scores to these declared endpoints; it does not establish that the formulas outperform every individual metric.

Table 1. Evaluation panels (six models each). COCO and VOC are used to select the frozen score formulas; all other panels are transfer evaluations. The three ADE20K panels share the same 2,000 validation images; the two street-scene panels share one checkpoint family.

Domain	Panel	Images	Models
General	COCO	4,031	FCN, DeepLabV3, LR-ASPP
	VOC	1,449	FCN, DeepLabV3, LR-ASPP
	ADE-S	2,000	SegFormer B0–B5
	ADE-X	2,000	UPerNet, BEiT-B, Mask2Former
	ADE-M	2,000	Mask2Former-L/B, OneFormer-L/T, UPerNet ConvNeXt-L, BEiT-L
Street	Cityscapes	500	SegFormer B0–B5 (Cityscapes)
	KITTI	200	SegFormer B0–B5 (Cityscapes)
CT	AMOS	100	TotalSegmentator (2), MONAI whole-body (2), Swin-UNETR, DiNTS

Within-panel comparisons use paired image-cluster bootstrap resampling of the same image groups: each draw carries all six model records of a sampled image together. Rank-endpoint draws resample jointly defined image-level paired differences; AUROC differences are recomputed exactly from the resampled records in every draw. One-sided empirical p-values are $(1 + \#\{\Delta^* \le 0\})/(B + 1)$ from 5,000 draws. The 40 panel-by-endpoint tests (eight panels by five endpoints) are Holm-adjusted together at $\alpha = 0.05$, including the eight mixed-failure identity comparisons. Intervals are conditional on the six observed checkpoints and fixed endpoint definitions; they do not quantify variation over future datasets or architectures. Aggregates across panels are descriptive, because panels are not independent draws from a defined population and the three ADE20K panels share images.

3.4. Domain-matched failure score

A failure score selected on one domain may not suit another whose failures differ in kind. To test whether matching the selection data to the target domain helps, we re-run the 260-formula selection for the mixed-failure endpoint on Cityscapes training records only (17,850 model–image records, of which 15,874 are severe misses and 2 are high-error failures), freeze the selected formula, and compare it with the general-scene mixed score on three held-out panels: Cityscapes validation, KITTI, and AMOS CT. Differences use the exact cluster bootstrap above with Holm correction over the three panels.

4. Panels and Protocol

General scenes. COCO val2017 [22] is evaluated through the 20 PASCAL foreground categories; only images with eligible components enter complete groups. All 1,449 VOC validation images [13] and 2,000 ADE20K validation

Table 2. Conflict bound B and the exact scalar minimum B_{exact} (all 720 orderings per image) on the eight panels.

	COCO	VOC	ADE-S	ADE-X	ADE-M	City	KITTI	CT
B	0.160	0.142	0.163	0.186	0.186	0.106	0.189	0.116
B_{exact}	0.160	0.143	0.164	0.187	0.187	0.107	0.191	0.116

images [34] are used. COCO and VOC use six torchvision models [6, 17, 25]. ADE-S contains SegFormer [32]; ADE-X contains UPerNet ConvNeXt and Swin [23, 24, 31], BEiT-B [2], and Mask2Former [8]; ADE-M contains recent Mask2Former, OneFormer [18], UPerNet ConvNeXt-L, and BEiT-L checkpoints, fixed before any was evaluated.

Street scenes. Cityscapes validation [10] and KITTI semantic segmentation [15] are evaluated with six Cityscapes-trained SegFormer checkpoints. The priority map assigns 5 to persons, riders, motorcycles, and bicycles and 4 to cars, trucks, buses, and trains. KITTI was not used to train any model.

Abdominal CT. We evaluate the 100 labelled validation CT volumes of AMOS [19] with six public models: TotalSegmentator [30] at full and fast resolution, MONAI [5] whole-body models at high and low resolution, Swin-UNETR [16] trained on BTCV, and a DiNTS multi-organ model, each run with its published pipeline. None is documented as trained on AMOS. Scoring uses the five organs all six models predict, with a clinically motivated priority map (pancreas 5; gallbladder and spleen 4; liver 3; stomach 2), components of at least 1 mL, and a three-dimensional port of the scorer that reproduces the two-dimensional one exactly.

Protocol. Class priorities, component eligibility (16 pixels in 2D), the high-error threshold, and the frozen formulas are fixed before each panel is evaluated. Checkpoint revisions are pinned, and re-running inference for the four original general-scene panels and Cityscapes reproduced all stored records exactly.

5. Results

5.1. Criterion conflict is unavoidable in every domain

The bound is positive on all eight panels (Table 2, Fig. 3), from 0.106 on Cityscapes to 0.189 on KITTI, and the exact minimum exceeds it by at most 0.0023. Intransitive preference cycles therefore add almost nothing: the conflict is pairwise and no scalar can remove it. On the general-scene panels, leave-one-criterion-out lower confidence limits are at least 0.114, counting ties as agreement lowers the bound to at least 0.123, splitting ties raises it to at most 0.261, and across all three-model subsets of ADE-M the bound ranges from 0.168 to 0.200. Its size depends on which criteria are declared: deployment-only criteria give 0.060–0.061 on ADE20K, versus 0.146–0.175 after adding the overlap criteria.

5.2. Endpoint-specific scoring transfers across domains

Scoring each endpoint with its own frozen formula improves policy-rank agreement, operational-rank agreement, and severe-miss AUROC on every panel (Fig. 4). All 24 of 24 contrasts are significant after Holm correction over all 40 panel-by-endpoint tests (largest adjusted $p = 0.0336$), and the smallest lower confidence limit is +0.011. The gains hold on panels never used for selection: recent ADE20K architectures (+0.207 in policy rank, +0.318 in severe-miss AUROC), street scenes (+0.341 and +0.175 on KITTI), and CT (+0.130 and +0.025).

Robustness. On the four original general-scene panels, re-running the full 260-formula selection on resampled COCO and VOC images in each of 1,000 bootstrap draws leaves 12 of 12 contrasts with intervals above zero, so the advantage does not depend on the particular formulas chosen. With the formulas frozen, the contrasts stay positive with intervals excluding zero under uniform priorities, all 202 single-class priority changes, and 100 random permutations of priority values. Recomputing AUROC exactly in each draw changes interval limits by at most 0.0019 relative to an influence-function approximation, and under a paired-exchange null the test rejects at 0.025–0.060 at nominal level 0.05.

Human judgments. To obtain an endpoint not built from the scores' own components, 4 reviewers, blind to models and scores, compared 100 pairs of real ADE-X predictions on which the endpoint-specific policy score and the mixed score disagree, answering one fixed question: which segmentation covers people and vehicles better. Protocol and analysis were pre-registered. On the 80 pairs with a clear majority, the majority sided with the endpoint-specific score on 61.3% (95% image-cluster bootstrap CI [50.0, 71.8]%; one-sided $p = 0.030$) and with the mixed score on 38.7%; inter-rater agreement was moderate (Fleiss' $\kappa = 0.50$). Among single criteria, policy utility matched the majority most often (70%) and false-positive quality least (43%). The effect is moderate, and on control pairs where both scores agree the majority matched them only 53% of the time, so the study supports the comparison between the two scores rather than close agreement of either with human judgment.

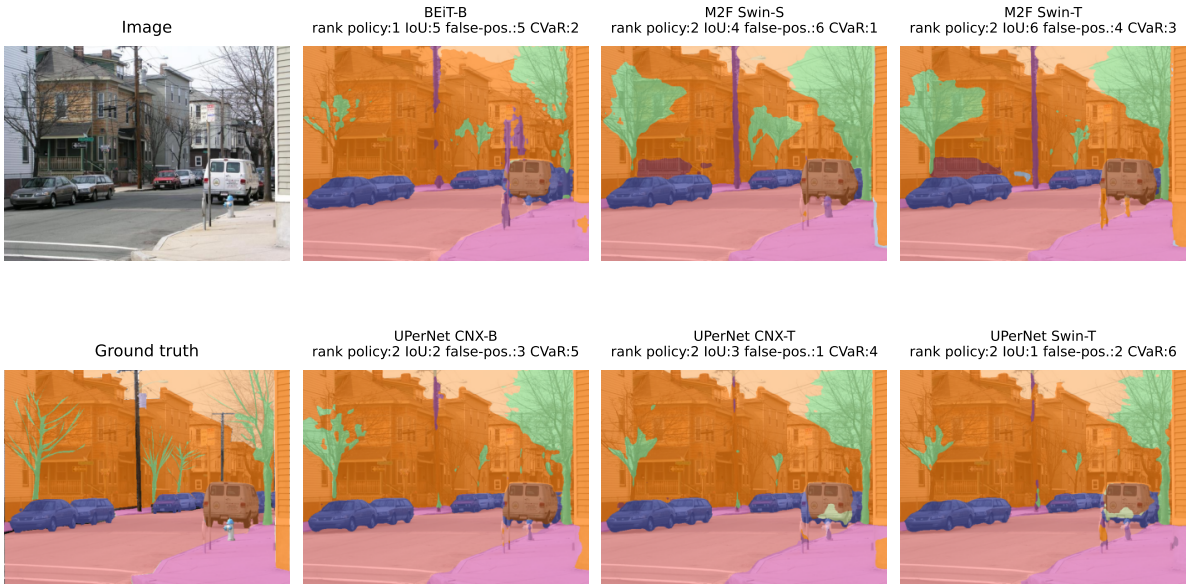


Figure 3. Real predictions of the six ADE-X models on one validation image, with each model's rank (1 = best; ties share a rank) under four of the seven criteria. The image was chosen by a fixed rule: among ADE-X images containing a critical component, with non-constant policy utility across models and at least 60 non-tied criterion votes, the one with the largest per-image exact conflict. No ordering of these six models agrees with all criteria.

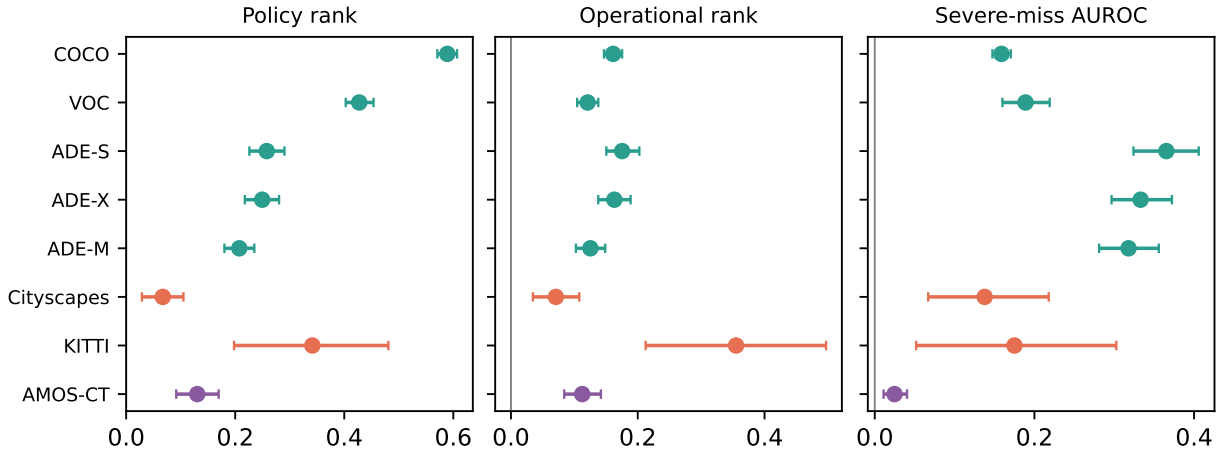


Figure 4. Endpoint-specific score minus reuse of the mixed score on all eight panels (95% paired image-cluster bootstrap intervals). Teal: general scenes; orange: street scenes; purple: abdominal CT. All contrasts remain significant after Holm correction over 40 tests.

5.3. A single failure score depends on failure composition

The general-scene mixed score, $h_{\text{mixed}} = 0.2\,q_{\text{comp}} + 0.8\,q_{\text{fp}}$, is the best failure detector on all five general-scene panels, with every interval above zero, and trails the strongest single metric on all three street-scene and CT panels, with every interval below zero (Table 3). The strongest single metric there is a coverage or tail score: severity arithmetic on Cityscapes and KITTI and CVaR on CT. The two groups differ in failure composition (Fig. 5): on general scenes high-error failures make up 58.4–63.2% of failures, on street scenes and CT at most 6.4%. The advantage of the mixed score also varies with the high-error threshold on general scenes, positive in 7/7 VOC cells, 5/7 COCO cells, and 4/7 cells on each of ADE-S and ADE-X.

In street scenes the residual failures are concentrated in small, distant objects. On Cityscapes 92.3% of model–image records contain a severe miss; in 69.0% of them the worst-

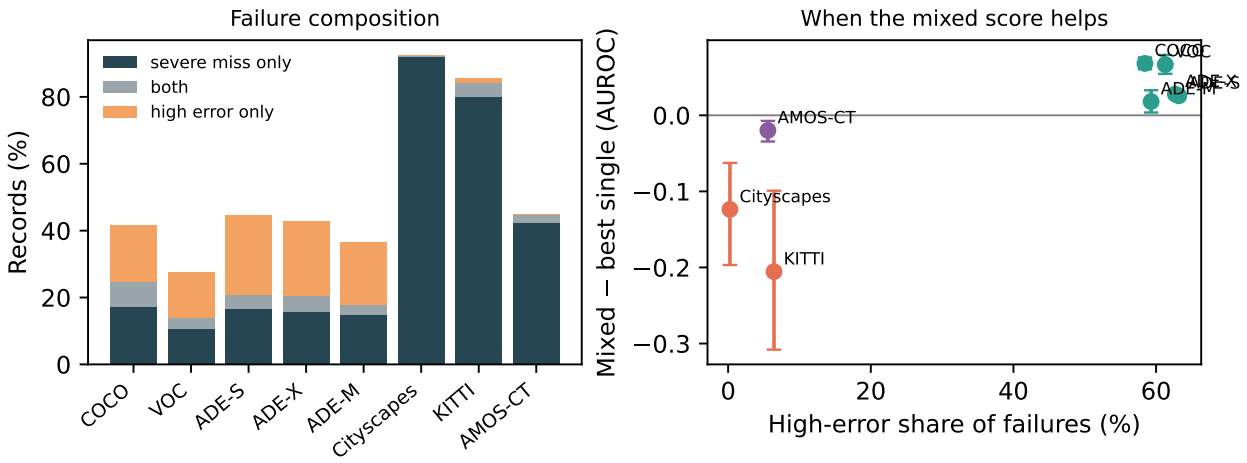


Figure 5. Left: composition of failure records on each panel. Right: frozen mixed failure score minus the strongest single metric (mixed-failure AUROC, 95% CI) against the share of failures involving high error.

Table 3. Mixed-failure AUROC. High: high-error share of failures. Frozen: general-scene mixed score minus the strongest single metric. Matched: score re-selected on Cityscapes training data, minus the frozen mixed score and minus the strongest single metric (95% CI).

Panel	High (%)	Frozen − best	Matched − frozen	Matched − best
COCO	58.4	+0.068 [+0.061, +0.076]	–	–
VOC	61.3	+0.067 [+0.054, +0.079]	–	–
ADE-S	63.2	+0.026 [+0.021, +0.030]	–	–
ADE-X	62.7	+0.028 [+0.023, +0.033]	–	–
ADE-M	59.3	+0.018 [+0.004, +0.033]	–	–
City	0.3	−0.124 [−0.197, −0.063]	+0.114 [+0.065, +0.169]	−0.010 [−0.028, +0.006]
KITTI	6.4	−0.206 [−0.308, −0.099]	+0.184 [+0.110, +0.260]	−0.022 [−0.060, +0.018]
CT	5.6	−0.020 [−0.034, −0.007]	+0.017 [+0.006, +0.034]	−0.002 [−0.009, +0.006]

covered critical component is a person, with median size 553 pixels in a 2,097,152-pixel image, and 67.9% of these components cover fewer than 1,024 pixels. A 16-pixel eligibility rule set for lower-resolution images admits these objects; as a sensitivity check only, ignoring critical components below 4,096 pixels lowers the severe-miss rate to 34.6%.

Domain-matched selection. Re-selecting the failure score on Cityscapes training records, in which severe misses vastly outnumber high-error failures, yields $0.4\,q_{\text{arith}} + 0.6\,q_{\text{fp}}$, where q_{arith} is severity-weighted arithmetic coverage. On the three held-out panels it beats the general-scene mixed score by +0.114 (Cityscapes validation), +0.184 (KITTI), and +0.017 (CT), all significant after Holm correction (largest adjusted $p = 0.0006$), and its gap to the strongest single metric is no longer distinguishable from zero (Table 3). Matching the selection data to the failure composition of the target domain thus restores the failure score, including on CT, a domain absent from its selection data.

6. Limitations

Shared construction. Policy utility, severe-miss labels, and the component scores are all built from component-coverage measurements. Agreement between an endpoint-specific score and its endpoint is therefore partly expected; the conflict bound and the cross-domain behaviour of a single failure score do not rely on this alignment. The human study (Sec. 5.2) provides one external check, but with four reviewers, one question, and a moderate effect; no downstream decision outcome was evaluated.

Analysis order. Formulas were selected on COCO and VOC and frozen before every transfer evaluation. The failure-composition analysis and the domain-matched selection were carried out after the street-scene and CT results had been observed. The domain-matched score was selected on Cityscapes training data only, but Cityscapes validation shares its source distribution; KITTI and CT provide the more independent test.

Panels. Each panel contains six models, and intervals are conditional on them. The three ADE20K panels share images, and the two street-scene panels share one Cityscapes-trained checkpoint family, so KITTI tests domain shift for that family rather than architecture diversity. The CT models differ widely in quality (per-model severe-miss rates 11%–91%), and part of one model family's training data is not publicly documented.

Endpoint definitions. Severity maps, including the clinically motivated CT map, are declared operationalizations, not validated harm weights. The fixed pixel threshold for component eligibility is resolution-dependent and can saturate severe-miss labels on high-resolution street imagery; eligibility rules should scale with image size.

Use in model selection. Endpoint-specific scoring requires the user to declare the decision objective; a misspecified objective is evaluated faithfully but wrongly. Conventional segmentation metrics, calibration, and cost should be reported alongside any endpoint-specific score.

7. Conclusion

Across eight six-model panels in general scenes, street scenes, and abdominal CT, seven declared segmentation criteria disagree in a way no scalar score can resolve: every ordering of the evaluated models contradicts at least 0.106–0.189 of the non-tied preferences, and exhaustive search shows this bound is essentially exact. Scoring each declared endpoint with its own frozen formula improved every non-trivial endpoint on every panel, including recent architectures and two domains never used for selection. A single failure score did not transfer from general scenes to domains whose failures are dominated by severe misses, and selecting it on data with matching failure composition restored it.

The practical implication is procedural rather than a new default metric. Evaluations should name their decision endpoint, state the composition of the failures they target, select any failure score on data whose failures resemble the target domain, scale component-eligibility rules with image resolution, and report criterion conflict rather than hide it inside a single aggregate.

References

[1] Kenneth J. Arrow. *Social Choice and Individual Values*. Wiley, New York, 1951.

[2] Hangbo Bao, Li Dong, Songhao Piao, and Furu Wei. BEiT: BERT pre-training of image transformers. In *International Conference on Learning Representations*, 2022.

[3] John Bartholdi, Craig A. Tovey, and Michael A. Trick. Voting schemes for which it can be difficult to tell who won the election. *Social Choice and Welfare*, 6(2):157–165, 1989.

[4] Jasmin Breitenstein, Florian Heidecker, Maria Lyssenko, Daniel Bogdoll, Maarten Bieshaar, J. Marius Zöllner, Bernhard Sick, and Tim Fingscheidt. What does really count? estimating relevance of corner cases for semantic segmentation in automated driving. In *Proceedings of the IEEE/CVF International Conference on Computer Vision Workshops*, pages 3991–4000, 2023.

[5] M. Jorge Cardoso et al. MONAI: An open-source framework for deep learning in healthcare. *arXiv preprint arXiv:2211.02701*, 2022.

[6] Liang-Chieh Chen, George Papandreou, Florian Schroff, and Hartwig Adam. Rethinking atrous convolution for semantic image segmentation. *arXiv preprint arXiv:1706.05587*, 2017.

[7] Bowen Cheng, Ross Girshick, Piotr Dollár, Alexander C. Berg, and Alexander Kirillov. Boundary IoU: Improving object-centric image segmentation evaluation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 15334–15342, 2021.

[8] Bowen Cheng, Ishan Misra, Alexander G. Schwing, Alexander Kirillov, and Rohit Girdhar. Masked-attention mask transformer for universal image segmentation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 1290–1299, 2022.

[9] Chih-Hong Cheng, Alois C. Knoll, and Hsuan-Cheng Liao. Safety metrics for semantic segmentation in autonomous driving. In *IEEE International Conference on Artificial Intelligence Testing*, pages 57–64, 2021.

[10] Marius Cordts, Mohamed Omran, Sebastian Ramos, Timo Rehfeld, Markus Enzweiler, Rodrigo Benenson, Uwe Franke, Stefan Roth, and Bernt Schiele. The Cityscapes dataset for semantic urban scene understanding. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pages 3213–3223, 2016.

[11] Pau de Jorge, Riccardo Volpi, Philip H. S. Torr, and Grégory Rogez. Reliability in semantic segmentation: Are we on the right track? In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 7173–7182, 2023.

[12] Cynthia Dwork, Ravi Kumar, Moni Naor, and D. Sivakumar. Rank aggregation methods for the web. In *Proceedings of the 10th International Conference on World Wide Web*, pages 613–622, 2001.

[13] Mark Everingham, Luc Van Gool, Christopher K. I. Williams, John Winn, and Andrew Zisserman. The pascal visual object classes (VOC) challenge. *International Journal of Computer Vision*, 88(2):303–338, 2010.

[14] Yonatan Geifman and Ran El-Yaniv. Selective classification for deep neural networks. In *Advances in Neural Information Processing Systems*, 2017.

[15] Andreas Geiger, Philip Lenz, Christoph Stiller, and Raquel Urtasun. Vision meets robotics: The KITTI dataset. *International Journal of Robotics Research*, 2013.

[16] Ali Hatamizadeh, Vishwesh Nath, Yucheng Tang, Dong Yang, Holger R. Roth, and Daguang Xu. Swin UNETR: Swin transformers for semantic segmentation of brain tumors in MRI images. In *International MICCAI Brainlesion Workshop*, 2022.

[17] Andrew Howard, Mark Sandler, Grace Chu, Liang-Chieh Chen, Bo Chen, Mingxing Tan, Weijun Wang, Yukun Zhu, Ruoming Pang, Vijay Vasudevan, et al. Searching for MobileNetV3. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 1314–1324, 2019.

[18] Jitesh Jain, Jiachen Li, MangTik Chiu, Ali Hassani, Nikita Orlov, and Humphrey Shi. OneFormer: One transformer to rule universal image segmentation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2023.

[19] Yuanfeng Ji, Haotian Bai, Chongjian Ge, Jie Yang, Ye Zhu, Ruimao Zhang, Zhen Li, Lingyan Zhang, Wanling Ma, Xiang Wan, and Ping Luo. AMOS: A large-scale abdominal multi-organ benchmark for versatile medical image segmentation. *Advances in Neural Information Processing Systems*, 2022.

[20] Kim-Celine Kahl, Carsten Lüth, Maximilian Zenk, Klaus Maier-Hein, and Paul F. Jäger. ValUES: A framework for systematic validation of uncertainty estimation in semantic segmentation. In *International Conference on Learning Representations*, 2024.

[21] John G. Kemeny. Mathematics without numbers. *Daedalus*, 88(4):577–591, 1959.

[22] Tsung-Yi Lin, Michael Maire, Serge Belongie, James Hays, Pietro Perona, Deva Ramanan, Piotr Dollár, and C. Lawrence Zitnick. Microsoft COCO: Common objects in context. In *European Conference on Computer Vision*, pages 740–755, 2014.

[23] Ze Liu, Yutong Lin, Yue Cao, Han Hu, Yixuan Wei, Zheng Zhang, Stephen Lin, and Baining Guo. Swin Transformer: Hierarchical vision transformer using shifted windows. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 10012–10022, 2021.

[24] Zhuang Liu, Hanzi Mao, Chao-Yuan Wu, Christoph Feichtenhofer, Trevor Darrell, and Saining Xie. A ConvNet for the 2020s. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 11976–11986, 2022.

[25] Jonathan Long, Evan Shelhamer, and Trevor Darrell. Fully convolutional networks for semantic segmentation. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pages 3431–3440, 2015.

[26] Lena Maier-Hein, Annika Reinke, Patrick Godau, Minu D. Tizabi, Florian Buettner, Evangelia Christodoulou, et al. Metrics reloaded: Recommendations for image analysis validation. *Nature Methods*, 21:195–212, 2024.

[27] Annika Reinke, Minu D. Tizabi, Michael Baumgartner, et al. Understanding metric-related pitfalls in image analysis validation. *Nature Methods*, 21:182–194, 2024.

[28] R. Tyrrell Rockafellar and Stanislav Uryasev. Optimization of conditional value-at-risk. *Journal of Risk*, 2(3):21–42, 2000.

[29] Furqan Ahmed Shaik, Abhishek Reddy, Nikhil Reddy Billa, Kunal Chaudhary, Sunny Manchanda, and Girish Varma. IDD-AW: A benchmark for safe and robust segmentation of drive scenes in unstructured traffic and adverse weather. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, pages 4614–4623, 2024.

[30] Jakob Wasserthal, Hanns-Christian Breit, Manfred T. Meyer, Maurice Pradella, Daniel Hinck, Alexander W. Sauter, Tobias Heye, Daniel T. Boll, Joshy Cyriac, Shan Yang, Michael Bach, and Martin Segeroth. TotalSegmentator: Robust segmentation of 104 anatomic structures in CT images. *Radiology: Artificial Intelligence*, 2023.

[31] Tete Xiao, Yingcheng Liu, Bolei Zhou, Yuning Jiang, and Jian Sun. Unified perceptual parsing for scene understanding. In *Proceedings of the European Conference on Computer Vision*, pages 418–434, 2018.

[32] Enze Xie, Wenhai Wang, Zhiding Yu, Anima Anandkumar, Jose M. Alvarez, and Ping Luo. SegFormer: Simple and efficient design for semantic segmentation with transformers. In *Advances in Neural Information Processing Systems*, pages 12077–12090, 2021.

[33] H. Peyton Young and Arthur Levenglick. A consistent extension of Condorcet's election principle. *SIAM Journal on Applied Mathematics*, 35(2):285–300, 1978.

[34] Bolei Zhou, Hang Zhao, Xavier Puig, Sanja Fidler, Adela Barriuso, and Antonio Torralba. Scene parsing through ADE20K dataset. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pages 633–641, 2017.